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ZS-MNET: A zero-shot learning based approach to multimodal named entity typing

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ABSTRACT

The task of named entity typing (NET) on social platforms is significant as it involves identifying the various types of named entities within unstructured text. The existing methods for NET only utilize the text modality to classify the types of named entities and ignore the semantic correlation of multimodal data. Moreover, the growing number of multimodal data implies a growing type set and the newly emerged entity types should be recognized without additional training. To address the aforementioned disadvantages, we introduce a zero-shot learning based multimodal NET (ZS-MNET) model that combines textual and visual modalities to recognize previously unseen named entity types in a zero-shot manner. The proposed ZS-MNET utilizes both text and image information to bridge the semantic correlation between multimodal data and label information, as opposed to the traditional zero-shot NET (ZS-NET) models. To incorporate fine-grained multimodal representations, we utilize pre-trained models that incorporate language and vision, particularly BERT and ViT, which are founded on transformer architectures. Besides, we propose the different multimodal representations to focus on fine-grained features for modeling semantic correlation between multimodal data and entity types in a fusion way. The experimental results underscore the utility of multimodal data in the NET field, while our approach surpasses previous ZS-NET models in performance.

1. Introduction

The assignment of semantic types to entity mentions, based on the local context of free text, is known as Named Entity Typing (NET), which serves as a crucial task. This task is highly effective and significantly contributes to downstream natural language processing (NLP) tasks, including question answering (Han, Kwon, Yu, & Lee, 2017), relation extraction (Liu et al., 2019), entity linking (Onoe & Durrett, 2020), and knowledge graph completion (Zhuo, Zhu, Yue, Zhao, & Han, 2022). The conventional NET methodologies concentrate on a collection of broadly defined categories and devise manually crafted attributes to deduce the types of named entity mentions (Chieu & Ng, 2002; Tjong Kim Sang & De Meulder, 2003). Considering to extend the coarse-grained types for the comprehensive definition of real-world entities, some researchers proposed fine-grained named entity typing (FNET) task which is concerned with fine-grained types organized in tree structure (Ling & Weld, 2012).

In real-world scenarios, there are ever-growing fine-grained entity types and it requires additional human effort to annotate each new type that emerges. To learn generalizable representations of named

entities and types without requiring additional annotation or training, zero-shot learning-based NET (ZS-NET) models were introduced. Ma, Cambria, and Gao (2016) employed a label embedding-based approach to leverage prototypical knowledge for inferring novel categories of named entities. To enhance the ZS-NET models' ability to concentrate on relevant information linked to entity types, the attention mechanism was incorporated (Obeidat, Fern, Shahbazi, & Tadepalli, 2019; Ren, Lin, & Zhou, 2020a). Zhang, Xia, Lu, and Yu (2020) introduced a memory-augmented model designed to utilize knowledge gained from known types for recognizing unseen ones. Besides, the multi-source auxiliary information of named entity types were exploited to enhance the ZS-NET model in a fusion way (Chen et al., 2021). The increasing amount of multimodal data available on social platforms is advantageous for NET in building multimodal knowledge graphs (Zhu et al., 2022). Hence, this paper introduces the Multimodal NET (MNET) task aimed at integrating textual and visual information for the identification of novel categories under the settings of zero-shot learning. Currently, the available ZS-NET techniques are ineffective at handling multimodal data. Because the previous methods are only focused on text modality

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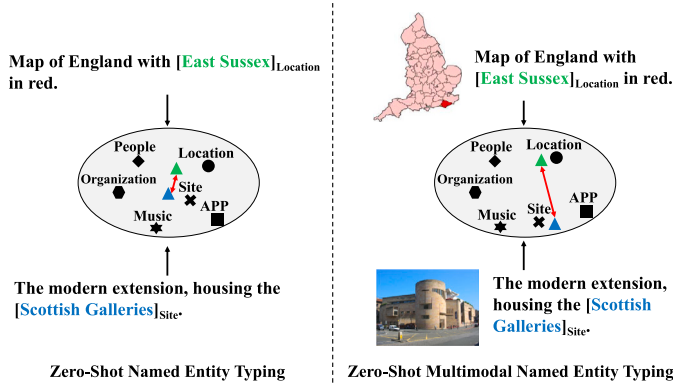


Fig. 1. The semantic space comparison between the zero-shot named entity typing (ZS-NET) and the zero-shot multimodal named entity typing (ZS-MNET).

and MNET needs to jointly model the multimodal semantic space for inferring unseen types of named entity mentions.

In order to address the aforementioned limitations of the current models, we introduce the zero-shot learning-based approach for multimodal named entity typing, referred to as ZS-MNET. The current ZS-NET methods acquire representations of entity mentions and their respective types within a collective semantic space, as demonstrated in Fig. 1. Despite their distinct semantic labels, the representations of entity mentions are too similar in the semantic space due to their shared context and type information, making it difficult to differentiate between them. We introduce ZS-MNET, a method that integrates text and image modalities to enhance the representation of entity mentions and types. This is in contrast to ZS-NET, as images possess semantic information that can augment the textual representations (Wan et al., 2021). Considering to exploit the fine-grained multimodal representations, we utilize the language and vision pre-trained models including: BERT (Devlin, Chang, Lee, & Toutanova, 2019) and ViT (Dosovitskiy et al., 2021) based on the transformer architectures (Vaswani et al., 2017). Furthermore, we propose different multimodal representation fusion modules to make ZS-MNET focus on fine-grained features for jointly modeling multimodal data and entity types in the shared semantic space. The manuscript's contributions can be summarized as follows:

1. We conclude that multimodal data holds significance for NET and introduce a new task named MNET in a zero-shot setting. To the best of our knowledge, we are the first to conduct an investigation of MNET within this specific context.
2. Taking into account the shortcomings of ZS-NET methods, we introduce a zero-shot learning approach for MNET that simultaneously represents multimodal data and entity types within a semantic framework.
3. The proposed model was compared to existing ZS-NET methods using a public dataset, and the experimental results demonstrated its superiority over the baseline approaches.

2. Related work

2.1. Zero-shot named entity typing

Ma et al. (2016) were the first to investigate fine-grained named entity typing using a zero-shot learning approach, known as ZS-NET. Compared with supervised NET, ZS-NET is defined to recognize the unseen types in the test set without additional training phase. Therefore, the vital challenge of ZS-NET lies in acquiring transferable representations of named entity mentions and types within the collective semantic space. Ma et al. (2016) proposed the label embedding model

to encode the prototypical information of types with word embeddings, and learn the semantic correlation between the entity mentions and types. Besides, Obeidat et al. (2019) employed the type descriptions sourced from Wikipedia to bolster the semantic distributed representations of various types. Considering to model the fine-grained semantic representations of entity mentions, Ren et al. (2020a) employed the attention mechanism to concentrate on the local features pertinent to the types. Zhang et al. (2020) designed the memory augmented ZS-NET model to keep seen types as memory components and transfer knowledge from seen types to unseen ones for explicitly modeling the semantic correlation between them. To leverage auxiliary information from various sources, Chen et al. (2021) introduced a multi-source fusion model that utilizes three distinct types of auxiliary data to enrich the representations of entity mentions and types.

The above studies investigate the ZS-NET based on the text modality, and we focus on ZS-MNET to make better use of multimodal data including text and image modalities. Unlike traditional ZS-NET with single modal learning, ZS-MNET is proposed to jointly model the semantic representations of text and image in a zero-shot manner.

2.2. Multimodal learning for information extraction

The traditional information extraction (IE) tasks, such as named entity recognition (Chieu & Ng, 2002) and entity linking (Schäfer, 2006), primarily concentrate on the textual modality. With the growing number of multimodal data, some researchers considered to model the rich semantic information of them and extended the above IE tasks to the multimodal IE ones which gained better results. Moon, Neves, and Carvalho (2018b) initially extended the traditional text-based named entity recognition to multimodal named entity recognition (MNER) and introduced the concept of modality attention module to integrate text information with image data for accurate sequence label prediction. In order to dynamically regulate the integration of text and image representations, Zhang, Fu, Liu, and Huang (2018) introduced an adaptive co-attention network, whereas Yu, Jiang, Yang, and Xia (2020) developed the uniform multimodal transformer. Furthermore, Xu, Huang, Sha, and Wang (2022) incorporated cross-modal alignment and matching modules into the MNER model to mitigate the impact of semantic mismatches between texts and images. Liu et al. (2024) suggested enhancing the textual representation through semantic augmentation and utilizing cross-modal semantic interaction across different visual granularities to achieve an effective multimodal guidance representation. In contrast to traditional text-based named entity linking, Moon, Neves, and Carvalho (2018a) were the first to introduce multimodal named entity linking (MNEL), which utilizes both textual and visual contexts to resolve entity disambiguation. Zhang, Li, and Yang (2021) applied the multiple attention mechanism on the MNEL for connecting the mention representation with its corresponding entity representation. Wang, Wu and Chen (2022) proposed a MNEL model with the gated hierarchical multimodal fusion and contrastive training for modeling the multimodal representations. To overcome the limitations of current MNEL datasets, which often feature restricted topics and entity types, Wang, Tian et al. (2022) introduced WikiDiverse, a publicly available dataset containing diverse entity types sourced from Wikinews articles covering various topics. Additionally, they employed multimodal pre-trained models to ensure consistent learning of textual and visual representations. Su, Xu, Yang, Chen, and Jing (2023) introduced a relational aggregation network designed to leverage the comprehensive details of an entity and its neighboring entities.

In essence, the multimodal IE models aim to strengthen the interaction between textual and visual modalities, thereby enhancing the semantic representations and facilitating their utility for various tasks. The above studies investigate the multimodal IE under the supervised setting, but we focus on the zero-shot MNET to recognize the unseen types without additional training phase. Unlike the supervised learning with adequate labeled data, the zero-shot learning is focused on modeling generalizable representations of samples and semantic labels for inferring samples with unseen types.

3. Model

We formalize the problem of zero-shot multimodal named entity typing prior to delving into the intricacies of the proposed model. The training set and the test set are represented by D_{tr} and D_{te} respectively. Within the training set, we establish a set of named entity types denoted as $\mathcal{Y}_s$ which encompasses $|\mathcal{Y}_s|$ distinct, observed types, expressed as $\mathcal{Y}_s = \{y_k^s\}_{k=1}^{|\mathcal{Y}_s|}$. The training dataset with $|D_{tr}|$ samples is formulated as $\{(S_i, V_i, l_i, r_i, y_i) | y_i \in \mathcal{Y}_s, i = 1, 2, \dots, |D_{tr}|\}$ where S_i and V_i are the sentence and image of i th sample. In addition, l_i and r_i represent the starting and ending indices of the entity mention within the sentence S_i , respectively, while y_i denotes the type label assigned to that mention. Under the requirement of zero-shot learning, the entity type set for test dataset D_{te} is defined as $\mathcal{Y}_u = \{y_k^u\}_{k=1}^{|\mathcal{Y}_u|}$ where $\mathcal{Y}_s \cap \mathcal{Y}_u = \emptyset$ and $|\mathcal{Y}_u|$ is the number of unseen types.

The zero-shot learning based model for multimodal named entity typing (ZS-MNET) is illustrated in Fig. 2. Besides of the language and vision pre-trained models for multimodal input representations, ZS-MNET is composed of the multimodal context representation, entity-aware and label-aware multimodal representations for jointly modeling semantic correlation between multimodal data and entity types.

3.1. Multimodal input representations

Given the multimodal data including sentences and images, we need to map them into the dense representations for neural networks. We define the input sentence with $|S|$ words as $S = \{w_1, w_2, \dots, w_{|S|}\}$. To leverage the outstanding performance of pre-trained language models, we employ BERT (Devlin et al., 2019) as the sentence encoder, transforming discrete words into distributed representations. Before being fed into BERT, each sentence requires the insertion of special tokens [CLS] and [SEP] at the beginning and end, respectively. And the pre-processed sentence is formulated as $\hat{S} = \{w_0, w_1, \dots, w_{|S|+1}\}$ where w_0 and $w_{|S|+1}$ represent the above special tokens respectively. The sentence representation calculation process can be simplified as $T = \text{BERT}(\hat{S})$ where $T = \{t_0, t_1, \dots, t_{|S|+1}\}$ and $t_i \in \mathbb{R}^d$ corresponding to the representation of token w_i . Furthermore, the indices indicating the beginning and conclusion of the entity mention within the sentence are designated as l and r respectively. Subsequently, the representation of the entity mention is computed by taking the average of the embeddings corresponding to the tokens within this range, formulated as $\mathbf{e} = \frac{1}{r-l+1} \sum_{i=l}^r t_i$.

For the visual information, we make use of ViT (Dosovitskiy et al., 2021) to extract the fine-grained representations of images. Compared with convolution neural networks (CNN) based pre-trained models like ResNet (He, Zhang, Ren, & Sun, 2016), ViT splits the image into small patches and uses transformer modules to keep the local vision information in the high-level representation. The representations of the image in the regional context are denoted by the set $\mathbf{V} = \{v_1, v_2, \dots, v_{|V|}\}$ where $v_i \in \mathbb{R}^d$ and $|V|$ represents the total number of feature vectors extracted from the ViT model.

3.2. Multimodal context representation

In order to capture the intricate semantic connections between sentences and images, the multimodal transformer is employed to derive a text-sensitive image representation. Unlike the text-based transformer (Vaswani et al., 2017), we regard the sentence representation $T \in \mathbb{R}^{d \times (|S|+2)}$ as queries, and the image representation $V \in \mathbb{R}^{d \times |V|}$ as key-value pairs. The multimodal transformer encompasses a multi-head attention (MHA) layer and a feed-forward neural network (FFNN). Initially, each individual head attention within the MHA is utilized to derive a text-sensitive image representation as

$$\mathbf{u}_i = \text{softmax}\left(\frac{[\mathbf{W}_{qi}T]^T[\mathbf{W}_{ki}V]}{\sqrt{d/m}}\right)[\mathbf{W}_{vi}V]^T \quad (1)$$

where $\{\mathbf{W}_{qi}, \mathbf{W}_{ki}, \mathbf{W}_{vi}\} \in \mathbb{R}^{d/m \times d}$ are the trainable weights for the i th head. The ultimate result of MHA with m attention heads is computed as $\mathbf{U} = \mathbf{W}'[\mathbf{u}_1; \mathbf{u}_2; \dots; \mathbf{u}_m]^T$, where $\mathbf{W}' \in \mathbb{R}^{d \times d}$ represents the weight matrix for MHA. Secondly, the application of residue connection on FFNN enables the calculation of the final output for the multimodal transformer, formulated as $\mathbf{F} = \text{LN}(\tilde{\mathbf{F}} + \text{FFNN}(\tilde{\mathbf{F}}))$ where $\tilde{\mathbf{F}} = \text{LN}(T + \mathbf{U})$ and LN is the layer normalization (Ba, Kiros, & Hinton, 2016).

Further more, we propose to model the multimodal context representation (MCR) by aggregating the semantic information of the two modalities in a gate way. To combine the sentence representation $T \in \mathbb{R}^{d \times (|S|+2)}$ and the text-aware image representation $\mathbf{F} \in \mathbb{R}^{d \times (|S|+2)}$ dynamically, we employ the gate mechanism to compute the representation of multimodal context as follows:

$$\mathbf{g}_1 = \text{sigmoid}(\mathbf{W}_{g1}[T; \mathbf{F}] + \mathbf{b}_{g1}), \quad (2a)$$

$$\mathbf{A} = [T; \mathbf{g}_1 \odot \mathbf{F}] \quad (2b)$$

where $\mathbf{W}_{g1} \in \mathbb{R}^{d \times 2d}$ and $\mathbf{b}_{g1} \in \mathbb{R}^d$ are the weights, $\odot$ denotes the element-wise production, and $\mathbf{A} = \{\mathbf{a}_0, \mathbf{a}_1, \dots, \mathbf{a}_{|S|+1}\}$ is the output representation of the multimodal context fusion module.

3.3. Entity-aware multimodal representation

In order to extract local representations of images that are related to specific entity mentions, we introduce an entity-aware gate fusion module. This module is designed to integrate multimodal representations, taking into account the semantic information associated with the mentioned entity. Given the entity mention representation $\mathbf{e} \in \mathbb{R}^d$ and image representation $\mathbf{V} \in \mathbb{R}^{d \times |V|}$, we employ the attention mechanism (Zhou et al., 2016) to derive the local representation of the image. The calculation of the entity-aware image representation is given by the formula $\mathbf{z} = \sum_{i=1}^{|V|} \alpha_i v_i$. And the attention score α is defined as $\alpha_i = \frac{\exp(\mathbf{W}_e[\mathbf{v}_i; \mathbf{e}])}{\sum_{i=1}^{|V|} \exp(\mathbf{W}_e[\mathbf{v}_i; \mathbf{e}])}$ where $\mathbf{W}_e \in \mathbb{R}^{2d}$ is the trainable parameter for

the attention module. Moreover, to fuse the semantic information of the entity mention and related regional image dynamically, we utilize the gate mechanism to model the entity-aware multimodal representation (EAMR). Therefore, given the entity mention representation $\mathbf{e} \in \mathbb{R}^d$ and entity-aware image representation $\mathbf{z} \in \mathbb{R}^d$, the entity-aware multimodal representation is calculated as follows:

$$\mathbf{g}_2 = \text{sigmoid}(\mathbf{W}_{g2}[\mathbf{e}; \mathbf{z}] + \mathbf{b}_{g2}), \quad (3a)$$

$$\mathbf{B} = [\mathbf{e}; \mathbf{g}_2 \odot \mathbf{z}] \quad (3b)$$

where the weight matrices $\mathbf{W}_{g2} \in \mathbb{R}^{d \times 2d}$ and $\mathbf{b}_{g2} \in \mathbb{R}^d$ along with the output representation $\mathbf{B} \in \mathbb{R}^{2d}$ constitute the entity-aware gate fusion module.

3.4. Label-aware multimodal representation

ZS-NET methods regarded the entity types as the prototypical information, and utilized them to learn the semantic correlation representations between the entity mentions and their corresponding types (Ma et al., 2016). To bridge the semantic gap between entity types and multimodal data, we introduce a label-aware multimodal context attention module that specifically targets fine-grained contextual representations semantically aligned with the type. Given the seen type set $\mathcal{Y}_s = \{y_1^s, y_2^s, \dots, y_{|\mathcal{Y}_s|}^s\}$ for training set, each entity type is defined as the readable word like: *location*. Therefore, we feed each entity type name into BERT with the same way to process sentences and apply the average pooling layer on the token sequences for obtaining the label embedding representation $\mathbf{l}_k \in \mathbb{R}^d$ of k th entity type in the seen type set. In order to derive precise contextual representations that are linked to specific entity types, we suggest developing label-aware multimodal representations (LAMR) and employing an attention mechanism to

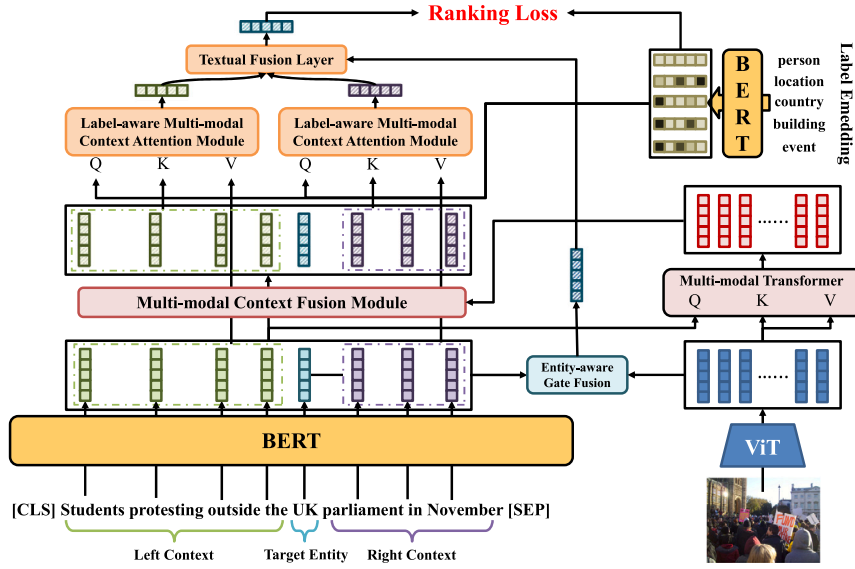


Fig. 2. The zero-shot learning based model for multimodal named entity typing (ZS-MNET).

determine the weight score as follows:

$$\beta_{k,i} = \frac{\exp(\mathbf{l}_k \mathbf{W}_{l1} \tanh(\mathbf{W}_{l2} \mathbf{A} + \mathbf{b}_{l2}))}{\sum_{i=1}^{|\mathcal{S}|+2} \exp(\mathbf{l}_k \mathbf{W}_{l1} \tanh(\mathbf{W}_{l2} \mathbf{A} + \mathbf{b}_{l2}))} \quad (4)$$

where $\mathbf{W}_{l1} \in \mathbb{R}^{d \times d}$, $\mathbf{W}_{l2} \in \mathbb{R}^{d \times 2d}$ and $\mathbf{b}_{l2} \in \mathbb{R}^d$ are the weight matrices. The calculations for the left and right context representations of an entity mention are given by the following equations: $\mathbf{c}_k^l = \sum_{i=1}^{l-1} \beta_{k,i} \mathbf{t}_i$ and $\mathbf{c}_k^r = \sum_{i=r+1}^{|\mathcal{S}|+2} \beta_{k,i} \mathbf{t}_i$ respectively. Further more, we combine the context information with EAMR as follows:

$$\tilde{\mathbf{c}}_k^l = \mathbf{W}_{l3}(\phi(\mathbf{c}_k^l; \theta_{l4}) \odot \phi(\mathbf{B}; \theta_{l5})) + \mathbf{b}_{l3}, \quad (5)$$

$$\tilde{\mathbf{c}}_k^r = \mathbf{W}_{l6}(\phi(\mathbf{c}_k^r; \theta_{l7}) \odot \phi(\mathbf{B}; \theta_{l8})) + \mathbf{b}_{l6} \quad (6)$$

where the weight matrices $\{\mathbf{W}_{l3}, \mathbf{W}_{l6}\} \in \mathbb{R}^{d \times d}$ and the bias vectors $\{\mathbf{b}_{l3}, \mathbf{b}_{l6}\} \in \mathbb{R}^d$ are employed in a fully connected layer activated by the nonlinear function $\tanh$, denoted as $\phi(\cdot; \theta)$. Therefore, the label-aware multimodal representations are denoted as follows:

$$\bar{\mathbf{l}}_k = [\mathbf{c}_k^l; \tilde{\mathbf{c}}_k^l; \mathbf{c}_k^r; \tilde{\mathbf{c}}_k^r], \quad (7a)$$

$$\bar{\mathbf{L}} = \{\bar{\mathbf{l}}_k\}_{k=1}^{|\mathcal{Y}_s|} \quad (7b)$$

where $\bar{\mathbf{L}} \in \mathbb{R}^{4d \times |\mathcal{Y}_s|}$ is the output LAMR of the label-aware multimodal context attention module.

3.5. Training and prediction procedure

For predicting the entity type, we regard the label embedding representations $\mathbf{L} = \{\mathbf{l}_k\}_{k=1}^{|\mathcal{Y}_s|} \in \mathbb{R}^{d \times |\mathcal{Y}_s|}$ as the prototypical information of entity types. Besides, we combine the EAMR with the LAMR by the textual fusion layer for obtaining the final representation $\bar{\mathbf{L}} = [\bar{\mathbf{B}}; \bar{\mathbf{L}}]$ where $\bar{\mathbf{B}} = [\mathbf{B}; \dots; \mathbf{B}] \in \mathbb{R}^{2d \times |\mathcal{Y}_s|}$ and $\bar{\mathbf{L}} = \{\bar{\mathbf{l}}_k\}_{k=1}^{|\mathcal{Y}_s|} \in \mathbb{R}^{6d \times |\mathcal{Y}_s|}$ is the output representation of ZS-MNET for the multimodal input data. To assess the association between the mentioned entity and its type, we employ a bilinear function to determine the score using the formula as $o_k = f(\mathbf{l}_k, \bar{\mathbf{l}}_k) = \bar{\mathbf{l}}_k^T \mathbf{W}_o \mathbf{l}_k$ where $\mathbf{W}_o \in \mathbb{R}^{6d \times d}$ and o_k is the relation score between the entity mention with the k th type in $\mathcal{Y}_s$.

Additionally, we leverage the ranking loss to ensure that the score of the true label remains higher than those of the false labels, and the objective of max-margin ranking is defined as follows:

$$\sum_{k=1, y_k^s \neq y}^{|\mathcal{Y}_s|} \max(0, 1 - o_{k'} + o_k) \quad (8)$$

where $y_{k'}^s = y$ is the true type for the entity mention. The overall procedure of ZS-MNET is illustrated in Algorithm 1. To update the

model weights, we employ stochastic gradient descent (SGD) techniques, incorporating multimodal data into the model and computing the loss using Eq. (8). During the prediction stage, we assess the scores of entity mentions in D_{te} against each unseen type in $\mathcal{Y}_u$, selecting the entity type with the highest score as the predicted type.

Algorithm 1 The overall procedure of ZS-MNET.

Input: The set of samples D and corresponding entity types $\mathcal{Y}$;

The initialized parameter of the overall ZS-MNET: Θ

- 1: **for** (S, V, l, r, y) in D **do**
- 2: /* Multimodal Input Representations */
- 3: $\mathbf{T}, \mathbf{V}, \mathbf{e} \leftarrow \text{MIR}(S, V, l, r)$
- 4: /* Multimodal Context Representation */
- 5: $\mathbf{A} \leftarrow \text{MCR}(\mathbf{T}, \mathbf{V})$
- 6: /* Entity-Aware Multimodal Representation */
- 7: $\mathbf{B} \leftarrow \text{EAMR}(\mathbf{e}, \mathbf{V})$
- 8: /* Label-Aware Multimodal Representation */
- 9: $\mathbf{L}, \bar{\mathbf{L}} \leftarrow \text{LAMR}(\mathcal{Y}, \mathbf{A}, \mathbf{T})$
- 10: /* Training and Prediction Procedure */
- 11: $\bar{\mathbf{L}} = \{\bar{\mathbf{l}}_k | \bar{\mathbf{l}}_k = [\mathbf{B}; \bar{\mathbf{l}}_k], \bar{\mathbf{l}}_k \in \bar{\mathbf{L}}, k = 1, 2, \dots, |\mathcal{Y}|\}$
- 12: $\mathbf{o} = \{o_k | o_k = f(\mathbf{l}_k, \bar{\mathbf{l}}_k), k = 1, 2, \dots, |\mathcal{Y}|\}$
- 13: $\hat{y} = \arg \max_{y_k} o_k$
- 14: Calculate loss by Eq. (8) and back-propagate it to update Θ
- 15: **end for**

Output: Trained parameter Θ , and prediction results $\hat{\mathcal{Y}} = \{\hat{y}_i\}_{i=1}^{|D|}$

4. Experiments

4.1. Datasets and experiment settings

The multimodal datasets including: Twitter (Adjali, Besançon, Ferret, Borgne, & Grau, 2020), Movie (Gan et al., 2021) and Weibo (Zhang et al., 2021) contain limited entity types like: *People* or *Organization*, and the Snap dataset (Moon et al., 2018a) is not public available. Therefore, we take the WikiDiverse (Wang, Tian et al., 2022) dataset as the benchmark one. Pairs of text and images, collected from Wikinews, comprise each sample in the dataset, with manual annotations indicating 13 distinct entity types within the sentences. The comprehensive statistical overview of the WikiDiverse dataset is presented in Table 1. To assess our model's performance in a zero-shot setting against baseline methods, we replicate this setup by dividing the WikiDiverse

Table 1

The statistical information of entity types in WikiDiverse dataset.

Entity type	# entities	# words	# sentences	Average length
People	4022	9087	3123	9.6
Other	2848	7632	2119	10.1
Location	2718	6592	2147	9.8
Organization	2261	6758	1952	10.2
Country	1743	4997	1535	9.7
Event	1141	3911	1040	11.1
Site	940	3442	864	10.5
Building	438	1745	420	10.1
Movie	68	414	62	10.7
Book	57	334	51	10.4
APP	42	231	38	9.5
Music	26	178	26	12.2
Currency	23	142	21	10.4

Table 2

The statistical information of entity types in the test set of MFIGER dataset.

Entity type	# entities	# words	# sentences	Average length
Location	79,788	141,002	74,561	30.5
Person	56,581	115,000	53,316	32.7
Organization	23,272	51,815	22,716	29.9
Other	10,334	42,167	9984	29.6
Event	7425	24,330	7335	30.2
Art	4003	19,501	3947	31.6
Product	3418	16,670	3263	31.1
Building	2027	10,551	2004	30.3

dataset's entity types into two categories: those observed during training and those encountered only during evaluation. Specifically, we keep the type *Other* as a challenging type and the remaining types are sorted by frequency (Aly, Vlachos, & McDonald, 2021). Then we assign the sorted entity types evenly between the three sets to keep the balance of the sample numbers as follows:

- $A = \{People, Site, Building, Currency\}$
- $B = \{Location, Event, Book, Music\}$
- $C = \{Organization, Country, APP, Movie\}$

To prevent bias in the distribution of type sets, we suggest assigning three distinct tasks to the training, validation, and test sets, respectively as follows:

- Task 1: train: A , validation: B , test: $C \cup \{Other\}$
- Task 2: train: B , validation: C , test: $A \cup \{Other\}$
- Task 3: train: C , validation: A , test: $B \cup \{Other\}$

For each task, we train the model using samples of the training set, and then fine-tune the hyper-parameters by assessing the model's performance on the samples of the validation set. Eventually, we report the evaluation metrics of the model on the samples with the type of test set.

Besides, we also exploited the test set of MFIGER dataset (Zhang et al., 2023) to demonstrate the generalization ability of ZS-MNET. The statistical information of the dataset is presented in Table 2. Given that the MFIGER dataset contains more samples than the WikiDiverse dataset and acquiring sufficient labeled training samples is impractical, we conduct an investigation to compare models under the cross-domain zero-shot setting.

We designate the WikiDiverse dataset as the source domain and the MFIGER dataset as the target domain. Using the training and validation sets of each above task, we train the model and then directly apply it to the test groups derived from the MFIGER dataset's test set for evaluation. To confirm that the entity type sets of training and test sets are disjoint with each other, we split the original MFIGER dataset into three groups as follows:

- Group 1: $\{Organization, Art, Product, Other\}$

- Group 2: $\{Person, Art, Product, Building, Other\}$
- Group 3: $\{Location, Art, Product, Event, Other\}$

The proposed model incorporates the fundamental versions of pre-trained language model BERT (Devlin et al., 2019) and vision model ViT (Dosovitskiy et al., 2021) to extract multimodal input representations. With a hidden layer size of 768, a dropout rate of 0.1, and a batch size of 4, the model is trained using a learning rate ranging from $[1e-5, 2e-5]$. During training, the model is initially trained on the training set for a maximum of 20 epochs before being tested on the validation set. Following an early stopping strategy, training ceases if the loss value on the validation set fails to decline over 10 consecutive epochs. Subsequently, the best-performing model is evaluated on the test set. All experiments benefit from the acceleration provided by NVIDIA GTX A6000 devices.

4.2. Compared methods

Given the absence of existing research on zero-shot multimodal named entity typing, we opt to compare the proposed model with previous zero-shot named entity typing (ZS-NET) methods. The prototype-driven label embedding method (Proto) was firstly proposed to encode prototypical information for ZS-NET (Ma et al., 2016). To focus on the local representations of sentences, the ZS-NET methods including: NZFET (Ren, Lin, & Zhou, 2020b) and DBZFET (Obeidat et al., 2019) utilized the attention mechanism to extract the fine-grained text representations correlated to the labels and the descriptions of them respectively. The memory augmented ZS-NET model (MZET) was proposed to transfer knowledge from seen types to unseen ones (Zhang et al., 2020). Besides, we utilize a prototype-based simple multimodal baseline (MMProto) which concatenates the entity mention and image representations based on BERT and ViT (Wan et al., 2021). Considering to exploit semantic information implied in multimodal data, the visual-context based network (MOVCNet) was proposed to tackle multimodal fine-grained entity typing (Zhang et al., 2023). Huang, Xu, Li, Ye, and Lin (2024) proposed the TPM-MI framework to utilize the visual representations as prompts for fusing textual ones. Zeng et al. (2024) utilized cross-modal transformers to design the ICKA framework for modeling multimodal representations and aligning multimodal knowledge.

4.3. Experimental results

On the WikiDiverse dataset, we conducted three tasks and evaluated ZS-MNET alongside baseline models. Given the varying sample sizes among different categories, we reported both micro-averaged F1 (F1-micro) and macro-averaged F1 (F1-macro) scores. Detailed experimental results are presented in Table 3. Our model excelled in most metrics, with average macro-averaged F1 and micro-averaged F1 scores surpassing the baselines by 5.0% and 10.1% respectively. Among text-based ZS-NET methods, DBZFET and NZFET consistently outperformed the Proto model, highlighting the importance of the attention module in capturing local sentence representations relevant to entity types. We propose ZS-MNET model to fuse multimodal data for recognizing unseen types and always achieve better performances than traditional text-based ZS-NET models demonstrating the superiority of the ZS-MNET. Although MMProto model also utilizes multimodal data for zero-shot multimodal named entity typing, it cannot gain better results than text-based methods. Because MMProto fuses multimodal data in a naive way and cannot capture the semantic correlation between entity types and multimodal information. Compared with MMProto, MOVCNet utilizes the cross-modal attention to extract the fine-grained visual representations corresponding to the textual ones and gains the better results than MMProto. But MOVCNet does not take the semantic information of entity types into account and it cannot achieve the better performance than our proposed model. Besides, the performances of each model on the three tasks are various which demonstrates that

Table 3

Performance comparison on the three tasks defined on the WikiDiverse dataset. The model's average scores across the three tasks indicate its overall performance.

Method	Task 1		Task 2		Task 3		Average	
	F1-macro	F1-micro	F1-macro	F1-micro	F1-macro	F1-micro	F1-macro	F1-micro
Proto	0.1501	0.2225	0.0581	0.0805	0.1991	0.2743	0.1358	0.1924
DBZFET	0.1159	0.1921	0.0705	0.1254	0.2627	0.4414	0.1497	0.2527
NZFET	0.1231	0.1951	0.1274	0.2128	0.2334	0.3130	0.1613	0.2403
MZET	0.1295	0.2043	0.1279	0.2056	0.1053	0.1827	0.1209	0.1975
MMPProto	0.1080	0.1333	0.0972	0.1356	0.1665	0.2364	0.1239	0.1684
MOVNet	0.1903	0.2942	0.1998	0.2905	0.1500	0.3583	0.1800	0.3143
TPM-MI	0.1223	0.2004	0.0790	0.1397	0.2543	0.4431	0.1518	0.2610
ICKA	0.1109	0.1707	0.0990	0.1601	0.1944	0.2940	0.1347	0.2082
ZS-MNET	0.1468	0.3084	0.2094	0.3039	0.2777	0.4691	0.2113	0.3605

Table 4

Performance comparison on the three groups defined on the MFINGER dataset under the cross-domain zero-shot setting.

Method	Group 1		Group 2		Group 3		Average	
	F1-macro	F1-micro	F1-macro	F1-micro	F1-macro	F1-micro	F1-macro	F1-micro
Proto	0.2264	0.3990	0.1779	0.4296	0.1349	0.2542	0.1797	0.3609
DBZFET	0.2052	0.2579	0.1611	0.2707	0.0871	0.0734	0.1511	0.2006
NZFET	0.3127	0.4008	0.1695	0.1658	0.0715	0.0669	0.1845	0.2111
MZET	0.1231	0.1149	0.1554	0.5030	0.2215	0.5442	0.1666	0.3873
MMPProto	0.2929	0.3414	0.2151	0.3562	0.0983	0.0999	0.2021	0.2658
MOVNet	0.2025	0.2088	0.1452	0.2752	0.1601	0.3503	0.1692	0.2781
TPM-MI	0.2606	0.3641	0.1572	0.1572	0.2912	0.6391	0.2363	0.3868
ICKA	0.3266	0.4841	0.1071	0.1683	0.1522	0.1576	0.1953	0.2700
ZS-MNET	0.3306	0.3911	0.2883	0.6831	0.2990	0.5711	0.3060	0.5484

different named entity typing models are sensitive to specific entity types.

Besides, we conduct the cross-domain zero-shot experiments with the use of MFINGER dataset to demonstrate the generalization ability of ZS-MNET as illustrated in Table 4. Our model outperformed in the majority of metrics, achieving average macro-averaged F1 and micro-averaged F1 scores that exceeded the baselines by 6.9% and 16.1%, respectively. Multimodal methods such as TPM-MI and ICKA yielded superior results compared to previous text-based approaches, highlighting the significance of multimodal information for cross-domain generalization capability. In essence, our model surpasses text-based and multimodal-based baselines due to our innovative approach of integrating text and image information, along with the development of sophisticated multimodal interaction modules that enhance multimodal and entity type representations, ultimately facilitating zero-shot named entity typing.

4.4. Further discussion

To investigate the model further, we conduct the analysis for presenting it in different aspects. We do the statistics of different entity type prediction results from ZS-MNET and NZFET to compare their performances in detailed. An ablation study has been conducted to validate the effectiveness of the various multimodal representations proposed in ZS-MNET. Furthermore, a case study was undertaken to explore the impact of multimodal data on named entity typing, and a visualization analysis was performed on the multimodal representation to demonstrate its utility.

4.4.1. Entity type prediction statistics

The performance of ZS-MNET and NZFET in predicting entity types is evaluated using a confusion matrix in Fig. 3. This matrix reveals several key insights into the strengths and weaknesses of these models. Firstly, ZS-MNET exhibits a strong bias towards predicting entity types that are more heavily weighted, such as *Other*, *People*, and *Location*. This suggests that the model is better at predicting entity types that are more prominent or common in the data. On the other hand, NZFET is more easily affected by the challenging type *Other*, indicating that it may struggle with less common or more ambiguous entity types.

Table 5

An ablation study was conducted on the WikiDiverse dataset for the ZS-MNET model, and the average scores represent the mean outcomes of the model across the three tasks.

Method	Average	
	F1-macro	F1-micro
ZS-MNET	0.2113	0.3605
w/o MCR	0.1440 (−0.0673)	0.1885 (−0.1720)
w/o EAMR	0.1773 (−0.0340)	0.2713 (−0.0892)
w/o LAMR	0.1797 (−0.0316)	0.2910 (−0.0695)

One possible explanation for the differences in performance between ZS-MNET and NZFET is the way they handle entity mentions with contexts. ZS-MNET enhances entity mention and type representations by integrating text information with imagery, enabling it to precisely differentiate between entity types sharing similar textual semantics but varying visual semantics. In contrast, NZFET employs entity mentions along with contextual information to bridge semantic gaps between them and entity types. However, this method may not be as adept at capturing subtle differences among various types of entity mentions.

Another theoretical insight from the confusion matrix is the similarity between entity mentions of different types. For example, the entity mentions of type *Site* are similar to those of type *People* in textual semantic level. This suggests that the models may struggle to distinguish between these types based solely on textual features, as the textual information may not be sufficient to capture the unique characteristics of each type.

4.4.2. Ablation study

To gain a comprehensive understanding of the efficacy of the multimodal representations proposed in ZS-MNET, an ablation study was conducted, the results of which are presented in Table 5. These findings reveal the crucial role played by each component in determining the overall performance of the model. Notably, the removal of the multimodal context representation (MCR) resulted in a significant drop in performance, highlighting the significance of integrating text and image low-level representations. This is because the low-level features from both modalities contain complementary information that

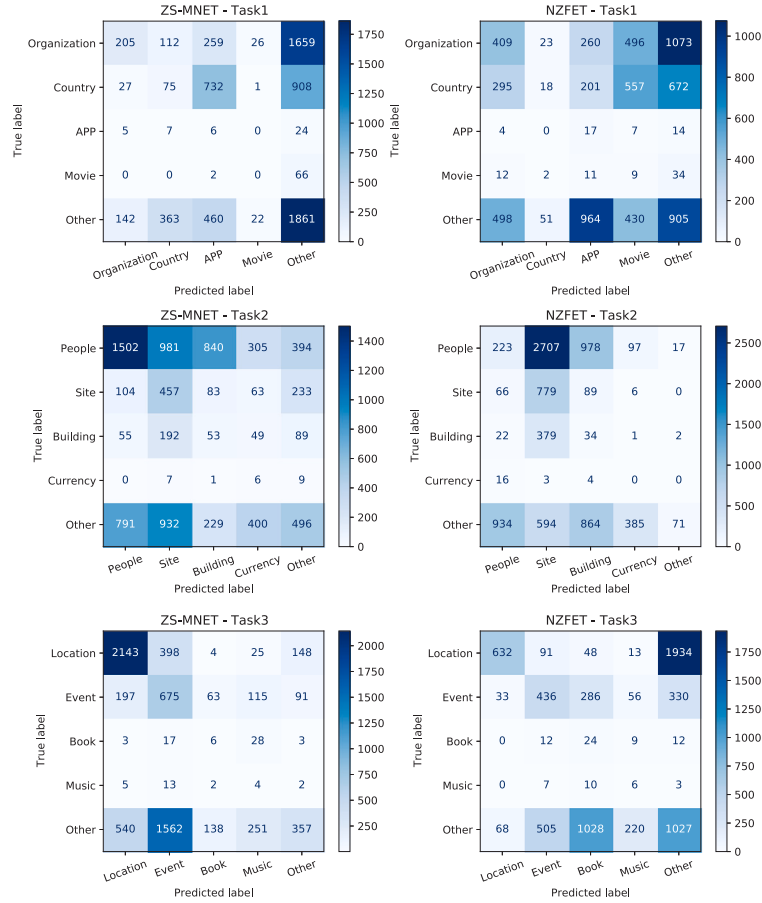


Fig. 3. The confusion matrix comparison between ZS-MNET and NZFET models on the test sets of the three tasks that are defined on the WikiDiverse dataset.

can enhance the overall understanding of the input data. By combining these features using MCR, the model can capture a richer set of contextual relationships between the modalities, leading to improved performance.

Secondly, we examine the contribution of entity-aware multimodal representation (EAMR) and label-aware multimodal representation (LAMR) to the final results. The results reveal that both EAMR and LAMR make significant contributions to the final performance of the model, albeit in different degrees. By employing a fusion approach to model fine-grained multimodal representations, EAMR enhances the model's ability to accurately capture semantic relationships among modalities. Conversely, LAMR, which integrates label information into multimodal representation, equips the model with the capability to effectively manage label data, thereby boosting overall performance.

The results obtained lend credence to the hypothesis that by fusing fine-grained multimodal representations, the model's performance can be enhanced. Specifically, the MCR and EAMR assist the model in capturing contextual relationships across modalities and semantic relationships within individual modalities, respectively. Furthermore, the integration of label information into the multimodal representation via LAMR enhances the model's ability to effectively process label information, thereby improving its overall performance.

4.4.3. Influence of pre-trained vision model

The pre-trained vision model encodes the image information into a dense representation for use in deep neural networks. To examine the impact of pre-trained vision models, we conducted experiments comparing the performance of ZS-MNET using various vision models, as illustrated in Fig. 4. During our experiments, we employed the base variant of ViT (Dosovitskiy et al., 2021), as the image encoder. Additionally, we chose the B32 (base) and L14 (large) configurations

Table 6

Two representative cases with manually labeled entity types were defined in the test sets based on the WikiDiverse dataset, along with the prediction outcomes obtained from various methods on these test samples.

Importance of Associated Image		Noise of Associated Image	
A. [Great Wall of China]Site 1805 map.		B. Hillary Clinton takes the [oath of office]Other.	
NZFET:	Building ✗	NZFET:	Other ✓
MMProto:	Site ✓	MMProto:	People ✗
ZS-MNET:	Site ✓	ZS-MNET:	People ✗

of ViT from CLIP (Radford et al., 2021) as our vision models for accomplishing the zero-shot named entity typing task. The CLIP-L14 model attained the highest macro-averaged F1 score, while the traditional ViT model obtained the best micro-averaged F1 score. The CLIP model was trained using contrastive learning on a vast amount of multimodal samples to align textual and visual representations. Despite being pre-trained solely on an image dataset, the traditional ViT was capable of capturing valuable visual representations. However, the base version of CLIP failed to yield superior results due to discarding a portion of visual information during the alignment of textual and visual representations. In conclusion, the performance of our proposed model can be influenced by various pre-trained vision models, and a larger vision model tends to yield better results.

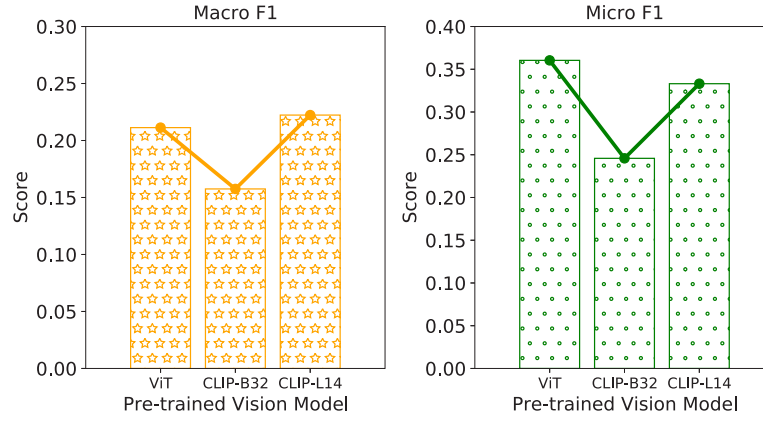


Fig. 4. The average performance comparison between different pre-trained vision models on the test sets of the three tasks that are defined on the WikiDiverse dataset.

Table 7

The representative cases sourced from the WikiDiverse dataset, along with the prediction results obtained from variants of ZS-MNET on these test samples.

Importance of the MCR module		Importance of the EAMR module		Importance of the LAMR module	
 A. Interior shot of the parliament's [Louise Weiss]Building building.		 B. Smoke coming from the wreckage at [San Francisco International Airport]Site.		 C. David Beckham at Annie Leibovitz's [Disney]Organization photo shoot in Spain.	
ZS-MNET:	Building ✓	ZS-MNET:	Site ✓	ZS-MNET:	Organization ✓
w/o MCR:	People ✗	w/o MCR:	Site ✓	w/o MCR:	Organization ✓
w/o EAMR:	Building ✓	w/o EAMR:	Other ✗	w/o EAMR:	Organization ✓
w/o LAMR:	Building ✓	w/o LAMR:	Site ✓	w/o LAMR:	Country ✗

4.4.4. Case study

The Table 6 displays a case study examining the influence of multimodal data on named entity typing. Specifically, Table 6.A demonstrates that the text-based NZFET model, relying solely on the entity mention with limited context, incorrectly predicts the entity type. Conversely, the multimodal-based ZS-MNET model effectively incorporates image information into the entity mention and type representations, enabling accurate entity type recognition. But the image information is not always beneficial to the named entity typing with the lack of understanding image scene semantics. As shown in Table 6.B, NZFET focuses on the text representations to recognize the entity type correctly, but ZS-MNET fuses the text information with image information to predict the type wrongly. The significant proportion of people in the image, along with the impact of semantic information on text representation, leads to prediction errors. Through evaluating the performance of text-based and multimodal models, we gain valuable insights into the difficulties and possibilities of integrating image data into named entity typing tasks.

Furthermore, as illustrated in Table 7, we have conducted a case study to explore the impact of the introduced MCR, EAMR, and LAMR modules on named entity typing. Specifically, Table 7.A showcases a sample featuring a building image, where the ZS-MNET, devoid of the MCR module, fails to predict the accurate entity type. This underscores the importance of low-level multimodal fusion representation, integrating textual and visual features, for ZS-MNET in identifying novel entity types where text and image exhibit semantic linkage. In Table 7.B, a sample depicting an airport image is presented, and the ZS-MNET without the EAMR module classifies the entity as *Other*. The EAMR module, however, effectively captures the semantic relationship between the textual entity representation and the visual airport image. Lastly, Table 7.C displays a sample with an image of David Beckham

and the sentence unrelated to the image. The ZS-MNET without the LAMR module misclassifies the entity. The LAMR module is tailored to capture the semantic correlation between the textual representation and the label representation for recognizing unseen types where text and image exhibit weak correlation.

4.4.5. Visualization analysis

To assess the efficacy of multimodal representations for named entity typing, we visualize the learned features extracted from ZS-MNET and NZFET in Fig. 5. We select entity mentions of specific types from the test sets and use ZS-MNET and NZFET to extract their representations, respectively. We then reduce the dimensionality of the output representations to two using t-SNE. The findings of our study demonstrate that the representations generated by NZFET exhibit clustering, indicating that the model has successfully learned distinct features for each entity type. On the other hand, the representations of ZS-MNET are more densely clustered within each type, indicating that multimodal data is more effective in capturing subtle differences between mentions of the same entity type.

This observation can be attributed to the unique properties of multimodal data. Textual semantic representations of entity mentions can vary greatly due to differences in syntax, semantics, and other contextual factors. In contrast, visual information tends to be more generalizable, as it is less susceptible to such variations and is more closely tied to the specific object features. By combining text and visual features, the multimodal data can better capture the inherent diversity of entity mentions, leading to more accurate and robust named entity typing. Furthermore, our findings suggest that the use of multimodal representations can mitigate the limitations of single-modal models, which are often prone to overfitting due to the inherent variability in textual data. By incorporating visual information, the model is able to

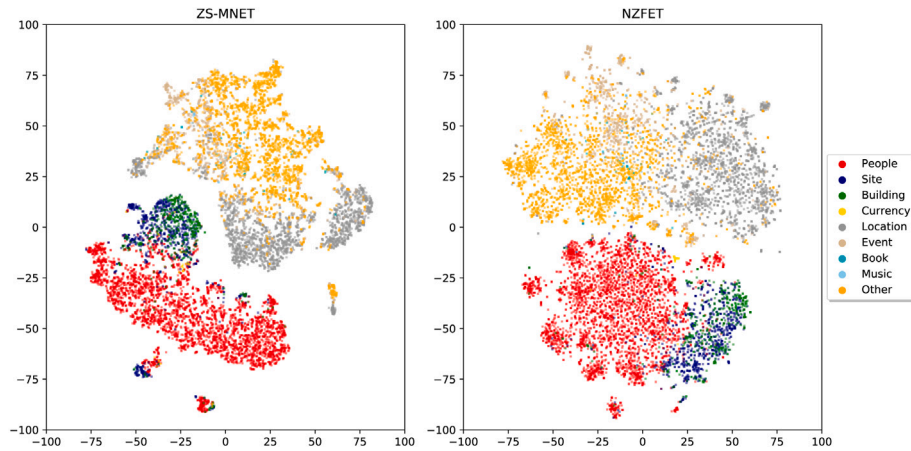


Fig. 5. Visualization of the t-SNE results exhibiting the representations of entities with particular types, which were extracted individually from ZS-MNET and NZFET.

Table 8

Comparison of average training and testing time (seconds for one sample) and the model size (MB).

Model	Train (s)	Test (s)	Size (MB)
MMPProto	0.1604	0.0751	744
MOVCNet	0.1779	0.0777	758
TPM-MI	0.1677	0.0768	755
ICKA	0.1764	0.0786	758
ZS-MNET	0.1793	0.0795	789

acquire more robust and transferable features, resulting in enhanced performance in named entity typing tasks.

4.4.6. Efficiency analysis

To perform an efficiency analysis of various multimodal models, we compare their time consumption and model sizes, as shown in Table 8. MMPProto, being the simplest model, boasts the smallest model size along with the minimum training and testing times. Despite having a larger model size and slightly longer training and testing times, existing multimodal methods such as MOVCNet, TPM-MI, and ICKA can outperform MMPProto in terms of results. When compared to multimodal baselines, ZS-MNET exhibits an average increase of 4.6% in model size and 4.0% in both training and testing times. Therefore, ZS-MNET is capable of achieving the best results in zero-shot multimodal named entity typing, albeit with a minor compromise in time efficiency.

5. Conclusion

This paper delves into the limitations of current zero-shot named entity typing techniques, primarily limited to textual data. To assess the utility of multimodal data in this domain, we introduce the concept of multimodal named entity typing, aiming to recognize novel types in a zero-shot manner. Additionally, we introduce a zero-shot learning-based framework for multimodal named entity typing (ZS-MNET), leveraging pre-trained language and vision models to fine-tune and fuse multimodal representations. Experimental findings reveal that ZS-MNET surpasses baseline methods, validating the significance of multimodal data in enhancing named entity typing across various dimensions. Our research offers fresh perspectives for future explorations in multimodal named entity typing.

In the future, we decide to develop methods that can better incorporate scene understanding and object relationship knowledge into the model architecture, or using techniques such as object detection and segmentation to identify and extract relevant image regions. Improving the model's ability to comprehend the semantics of image scenes can lead to reduced prediction errors in ZS-MNET models.

CRediT authorship contribution statement

Baohang Zhou: Writing – original draft, Software, Methodology, Conceptualization. **Ying Zhang:** Writing – review & editing, Project administration, Funding acquisition. **Kehui Song:** Writing – review & editing, Data curation. **Xuhui Sui:** Writing – review & editing, Visualization, Data curation. **Yu Zhao:** Writing – review & editing, Software, Resources. **Xiaojie Yuan:** Supervision, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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